

Financial Analysis Project in Excel

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1. Introduction

This project provides hands-on insights into Excel's financial function toolkit, covering excel **functions** across loan analysis, investment valuation, and return calculation. By applying these functions to structured datasets, the project develops the ability to make evidence-based financial decisions using Excel.

Core Objectives

- Calculate present value, net present value, and internal rate of return
- Build a full EMI amortisation schedule with IPMT, PPMT, CUMIPMT and CUMPRINC
- Determine effective interest rate and loan term using RATE and NPER
- Compare investments using NPV/XNPV and evaluate returns via IRR/XIRR/MIRR
- Produce a professional summary report suitable for interview presentation

2. Project Scope & Objectives

Excel-Based Financial Analysis	Utilise Microsoft Excel to perform comprehensive financial analysis using 13 built-in financial functions.
Real-World Problem Solving	Apply functions to genuine scenarios: refrigerator purchase decisions, home loan scheduling, investment comparison, and project return evaluation.
Professional Summary Report	Create a polished report covering methodology, findings, and actionable insights — ready for interview.

3. Complete Function Reference — All 13 Functions

#	Function	Category	Purpose	Excel Syntax
1	PV	Valuation	Present value of future payments	=PV(rate, nper, pmt, [fv], [type])
2	NPV	Valuation	Net present value at a discount rate	=NPV(rate, value1, [value2], ...)
3	XNPV	Valuation	NPV with irregular cash flow dates	=XNPV(rate, values, dates)
4	PMT	Loan	Fixed periodic payment / EMI	=PMT(rate, nper, pv, [fv], [type])
5	IPMT	Loan	Interest portion of one payment	=IPMT(rate, per, nper, pv, [fv], [type])
6	PPMT	Loan	Principal portion of one payment	=PPMT(rate, per, nper, pv, [fv], [type])
7	CUMIPMT	Loan	Cumulative interest between two periods	=CUMIPMT(rate, nper, pv, start, end, type)
8	CUMPRINC	Loan	Cumulative principal between two periods	=CUMPRINC(rate, nper, pv, start, end, type)
9	RATE	Loan	Interest rate per period (back-solve)	=RATE(nper, pmt, pv, [fv], [type], [guess])
10	NPER	Loan	Number of payment periods (back-solve)	=NPER(rate, pmt, pv, [fv], [type])
11	IRR	Return	Internal rate of return (periodic)	=IRR(values, [guess])
12	XIRR	Return	IRR with irregular cash flow dates	=XIRR(values, dates, [guess])
13	MIRR	Return	Modified IRR — separate finance & reinvest rates	=MIRR(values, finance_rate, reinvest_rate)

■ **Valuation** (PV, NPV, XNPV)
 ■ **Loan** (PMT, IPMT, PPMT, CUMIPMT, CUMPRINC, RATE, NPER)
 ■ **Return** (IRR, XIRR, MIRR)

Function-by-Function Explanation

PV — Present Value

The current worth of a stream of equal future payments, discounted at a given rate. **type=0** = payments at end of period (ordinary annuity); **type=1** = beginning of period (annuity due). A lower PV means a cheaper deal for the same nominal payments.

NPV — Net Present Value

Sums the present value of all cash flows (positive and negative) at a chosen discount rate. NPV assumes cash flows occur at the **end** of each period. **NPV > 0**: investment adds value. **NPV < 0**: investment destroys value. Note: the initial investment is usually added separately since NPV treats value1 as period-1 inflow.

XNPV — Extended NPV

Handles cash flows occurring on specific, irregular dates — the norm in real projects. The first date must be the earliest. Unlike NPV, it does not assume equal time intervals. Formula: $\text{sum of cashflow}_i / (1 + \text{rate})^{((\text{date}_i - \text{date}_0) / 365)}$.

PMT — Payment / EMI

Computes the fixed periodic payment (EMI) needed to fully repay a loan. For a monthly EMI: use $\text{rate}/12$ and $\text{nper} = \text{years} \times 12$. PV = loan amount; FV = 0 (loan fully paid off); type = 1 for beginning-of-month payments.

IPMT — Interest Component

Returns the interest portion of a specific payment number (per). As the loan progresses, IPMT decreases since the outstanding balance shrinks — early payments are predominantly interest.

PPMT — Principal Component

Returns the principal repaid in a specific payment. PPMT increases over time as IPMT decreases. Verification: $\text{IPMT} + \text{PPMT} = \text{PMT}$ for every period.

CUMIPMT — Cumulative Interest

Calculates the total interest paid between two payment periods (start_period to end_period), inclusive. Saves building a full amortisation table just to know total interest cost over a date range. **type=0** = end-of-period payments; **type=1** = beginning-of-period.

CUMPRINC — Cumulative Principal

Calculates the total principal repaid between two payment periods, inclusive. Together with CUMIPMT it gives full visibility into any loan slice — e.g. "How much principal have I repaid in years 2–5 of my mortgage?"

RATE — Interest Rate

Back-solves for the periodic interest rate given the number of periods, payment, and present value. Returns the rate per period — multiply by 12 for annual rate if using monthly inputs. Use the optional [guess] argument (~0.1) if RATE returns an error.

NPER — Number of Periods

Back-solves for how many payment periods are required to repay a loan. Useful for answering: "If I can afford ■ X/month, how long will it take to clear ■ Y at Z% interest?"

IRR — Internal Rate of Return

The discount rate at which NPV = 0 — the investment's break-even rate. Use the [guess] parameter and vary it to check for uniqueness. Returns #NUM! if no IRR exists. Assumes equally spaced, periodic cash flows.

XIRR — Extended IRR

IRR for irregularly dated cash flows. Requires a date range alongside cash flows. More accurate than IRR for real investments where payments do not follow a fixed calendar.

MIRR — Modified IRR

Addresses two IRR weaknesses: (1) the unrealistic assumption that intermediate cash flows are reinvested at the IRR itself, and (2) multiple IRR solutions. MIRR accepts a separate `finance_rate` (cost of capital) and `reinvest_rate` (rate on reinvested cash), always producing a single, interpretable result. **Preferred over IRR in professional practice.**

4. Implementation & Worked Examples

Example 1 — PV: Refrigerator Purchase Decision

A refrigerator costs ■32,000. Option: pay ■6,000/year for 8 years at 13% p.a. Which option is cheaper in today's money?

Option	Payment Structure	Excel Formula	Present Value	Verdict
1	Pay ■32,000 now	—	■ 32,000	
2	■ 6,000/yr — end of year	<code>=PV(13%, 8, -6000)</code>	■ 28,793	✓ Cheapest
3	■ 6,000/yr — beginning of year	<code>=PV(13%, 8, -6000,,1)</code>	■ 32,536	

Insight: Option 2 (end-of-year payments) costs only ■28,793 in present value — ■3,207 less than paying upfront. Always compare PV, not nominal totals.

Example 2 — PMT / IPMT / PPMT / CUMIPMT / CUMPRINC: Loan Amortisation

Loan: ■1,00,000 | Rate: 16% p.a. (1.333%/month) | Term: 8 months | Type: 0 (end-of-period)

Step 1 — Calculate EMI using PMT:

`=PMT(16%/12, 8, -100000)` → EMI = ■13,242.27

Step 2 — Amortisation schedule (IPMT & PPMT per month):

Mon th	Opening Balance	EMI	Interest (IPMT)	Principal (PPMT)	Closing Balance
1	■ 1,00,000.00	■ 13,242.27	■ 1,300.00	■ 11,942.27	■ 88,057.73
2	■ 88,057.73	■ 13,242.27	■ 1,144.75	■ 12,097.52	■ 75,960.22
3	■ 75,960.22	■ 13,242.27	■ 987.48	■ 12,254.78	■ 63,705.43
4	■ 63,705.43	■ 13,242.27	■ 828.17	■ 12,414.10	■ 51,291.34
5	■ 51,291.34	■ 13,242.27	■ 666.79	■ 12,575.48	■ 38,715.86
6	■ 38,715.86	■ 13,242.27	■ 503.31	■ 12,738.96	■ 25,976.89
7	■ 25,976.89	■ 13,242.27	■ 337.70	■ 12,904.57	■ 13,072.33
8	■ 13,072.33	■ 13,242.27	■ 169.94	■ 13,072.33	■ 0.00 ✓

Step 3 — Cumulative amounts between Month 2 and Month 3 (CUMIPMT & CUMPRINC):

Function	Formula	Result	Verification (Sum of individual values)
CUMIPMT	<code>=CUMIPMT(16%/12, 8, 100000, 2, 3, 0)</code>	■ 2,132.23	■1,144.75 + ■987.48 = ■2,132.23 ✓
CUMPRINC	<code>=CUMPRINC(16%/12, 8, 100000, 2, 3, 0)</code>	■ 24,352.30	■12,097.52 + ■12,254.78 = ■24,352.30 ✓

Key observation: Interest decreases every month (shrinking balance) while principal increases — this is amortisation. CUMIPMT and CUMPRINC let you instantly total interest/principal over any date range without building a full schedule. Results match the row-by-row verification exactly.

Example 3 — NPV / XNPV: Investment Comparison

Discount rate: 20% p.a.

Year	Investment 1 Cash Flow	Investment 2 Cash Flow
1	–■10,000	–■5,000
2	■25,000	■20,000
3	–■7,000	–■8,000
Nominal Total	■8,000	■7,000
NPV (End-Year)	■4,976.85	■5,092.59 ✓ Better
NPV (Beg.-Year)	■5,972.22	■6,111.11 ✓ Better
NPV (Mid-Year)	■5,451.87	■5,578.66 ✓ Better

Insight: Investment 1 has a higher raw total (■8,000 vs ■7,000) yet Investment 2 is superior across all NPV variants because its cash outflows occur later — later payments cost less in present value. For mid-year cash flows, multiply end-year NPV by $\sqrt[3]{(1+r)}$.

XNPV usage: When cash flows fall on specific calendar dates (not annual periods), use =XNPV(0.2, cashflow_range, dates_range). The first date must be the earliest in the dataset.

Example 4 — RATE & NPER: Back-Solving Loan Parameters

Question	Given	Formula	Answer
What interest rate am I paying?	Loan: ■1,00,000 EMI: ■12,000 mo Term: 15 months	=RATE(15, -12000, 100000)	8% per month
How long to repay?	Loan: ■1,00,000 EMI: ■15,000 mo Rate: 10% p.a.	=NPER(10%/12, -15000, 100000)	12 months

Tip: RATE returns the periodic rate. Multiply by 12 to annualise for monthly inputs. If RATE returns #NUM!, provide a [guess] value such as 0.1 (10%). NPER may return a decimal — always round up to ensure the loan is fully repaid.

Example 5 — IRR / XIRR / MIRR: Project Return Analysis

Function	Cash Flows / Input	Formula	Result	Notes
IRR (unique)	■10,000 –■5,000 –■8,500 ■2,000	=IRR(B3:B6)	10.53%	Unique — constant across all guess values
IRR (multiple)	–■20,000 ■82,000 –■60,000 ■2,000	=IRR(B3:B6, 0.15) =IRR(B3:B6, 0.45)	–9.59% & 216.09%	Two sign changes → two IRRs; use MIRR instead
XIRR	–■10,000 (8 Apr 2015) ■4,000 / ■3,000 / ■5,000 (Aug 2015 / Mar 2016 / Apr 2016)	=XIRR(C3:C6, B3:B6)	26.42%	Irregular dates — XIRR required

MIRR	Finance rate: 10% Reinvest rate: 12% –■1.6 ■10 –■10	=MIRR (D6:D8, D2, D3)	7%	Single result even with multiple IRRs ;
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IRR Rule: One sign change in cash flows → unique IRR guaranteed. More than one sign change → IRR may be multiple or not exist (#NUM!). In such cases, use MIRR for a single, reliable answer.

NPV vs IRR conflict (mutually exclusive projects): IRR may rank a smaller project higher due to scale differences. **Always use NPV as the final decision criterion.**

5. Key Findings & Insights

1 Time Value of Money drives everything

The same nominal amount is worth more today than in the future. Every function in this project is a direct application of this single principle. Payment timing (beginning vs end of period) can change present value by thousands of rupees.

2 NPV is the gold-standard investment metric

Raw totals and even IRR can mislead when projects differ in scale or cash flow timing. NPV correctly values every cash flow at the appropriate discount rate and always gives a reliable basis for comparison.

3 Amortisation structure: interest-heavy early, principal-heavy late

In the 8-month loan example, Month 1 interest is ■1,300 but falls to ■170 by Month 8. IPMT and PPMT expose this structure; CUMIPMT and CUMPRINC let you instantly total any range.

4 IRR has three scenarios — know them all

A unique IRR (one sign change) is safe to use. Multiple IRRs (multiple sign changes) require MIRR. No IRR (#NUM! for all guesses) means the project has no break-even rate and NPV must decide.

5 MIRR is superior to IRR in professional practice

MIRR corrects the reinvestment-at-IRR assumption and always produces a single answer. In the project example, MIRR returned 7% where IRR was ambiguous.

6 Use X-variants for real-world irregular cash flows

XNPV and XIRR are the professional standard when cash flows don't follow a fixed calendar. The XIRR example produced 26.42% — a figure that standard IRR could not calculate at all.

7 RATE and NPER complete the loan toolkit

Beyond calculating EMI, Excel can back-solve: what rate am I paying? (RATE) and how long will repayment take? (NPER). These are essential for real-world loan negotiation scenarios.

6. Decision-Making Framework

Choose the right function for the right question:

Scenario / Question	Function	Why
How much is this future income stream worth today?	PV	Discounts regular equal payments
Is this investment profitable (regular intervals)?	NPV	Sums all discounted cash flows
Is this investment profitable (irregular dates)?	XNPV	Date-aware NPV
What is my monthly loan repayment?	PMT	Fixed instalment from loan terms
How much interest am I paying in month N?	IPMT	Interest component per period
How much principal am I repaying in month N?	PPMT	Principal component per period
Total interest paid between months A and B?	CUMIPMT	Cumulative interest over a range
Total principal repaid between months A and B?	CUMPRINC	Cumulative principal over a range
What interest rate am I being charged?	RATE	Back-solves for rate
How many months to repay?	NPER	Back-solves for periods
What return does this investment generate (periodic)?	IRR	Break-even discount rate
What return — cash flows on irregular dates?	XIRR	Date-aware IRR
Finance rate $\neq$ reinvest rate, or multiple IRRs?	MIRR	Single result, realistic assumption

■ **NPV vs IRR Conflict:** For mutually exclusive projects, IRR can give wrong rankings when projects differ in scale or cash flow timing. A small project with 50% IRR may be inferior to a large project with 17% IRR if the NPV of the latter is higher. **Always use NPV as the final decision criterion.**

7. Conclusions & Recommendations

1. All 13 Excel financial functions (PV, NPV, XNPV, PMT, IPMT, PPMT, CUMIPMT, CUMPRINC, RATE, NPER, IRR, XIRR, MIRR) are production-ready tools used by financial analysts daily.
2. The time value of money is the foundational principle behind every function in this project.
3. NPV is the gold-standard metric for investment decisions — more reliable than IRR alone.
4. CUMIPMT and CUMPRINC give instant transparency over total loan costs across any period range.
5. MIRR and XIRR are the professional-grade alternatives to IRR — preferred in real-world practice.
6. RATE and NPER complete the loan toolkit, enabling back-solving for unknown parameters.
7. For irregular cash flows (the norm in practice), always use XNPV and XIRR over their periodic counterparts.

Project Summary

This Financial Analysis Project in Excel delivered a comprehensive, hands-on mastery of 13 core Excel financial functions. Through five structured worked examples — from a simple purchase decision to complex irregular-date investment analysis — the project demonstrates the ability to translate financial theory into precise, Excel-based analytical solutions that support real-world decision-making.
